SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	D. Sri Anjaneya	**Reg. No.:**	192425219
Section / Batch:	AI & ML / 2024	**Date:**	30/07/2026

Code of Conduct

I D. Sri Anjaneya / 192425219 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: D. Sri Anjaneya

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3

Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.
Apply rules to decompose each word into root and suffix components.
Differentiate between inflectional and derivational forms within the dataset.
Normalize all variants to a common base form for efficient retrieval.
Determine the parsed structure and normalized output for each word.

2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.
Identify prefixes, suffixes, and base forms using rule-based decomposition.
Ensure semantic consistency across derived word forms.
Classify each transformation as inflectional or derivational.
Produce the root and morphological breakdown for each word.

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.
Apply morphological rules to strip affixes and identify the base form.
Handle both grammatical variations and word formation changes.
Ensure consistent root extraction across all forms.
Determine the stem and classification for each input word.

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.
Apply state transitions to represent suffix transformations and irregular forms in verb inflection.
Differentiate between regular and irregular inflectional patterns within the dataset.
Ensure consistent mapping of all forms to a common root representation.
Determine the state transitions, morphological breakdown, and normalized output for each word.

5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.
Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.
Handle variations in suffix patterns while preserving semantic consistency.
Ensure uniform root extraction across all derived forms.
Determine the stemming steps, intermediate forms, and final stem for each input word.

1. Morphological Analysis for Document Indexing System

Word 1: connected

Rule Applied: Remove past tense suffix -ed

connected
→ connect + ed
→ remove ed
→ connect

Root: connect

Suffix: -ed

Type: Inflectional

Normalized Output: connect

Word 2: connecting

Rule Applied: Remove present participle suffix -ing

connecting
→ connect + ing
→ remove ing
→ connect

Root: connect

Suffix: -ing

Type: Inflectional

Normalized Output: connect

Word 3: connection

Rule Applied: Remove derivational suffix -ion

connection

→ connect + ion

→ remove ion

→ connect

Root: connect

Suffix: -ion

Type: Derivational

Normalized Output: connect

Final Parsed Structure

connected → [connect] + [-ed] → connect

connecting → [connect] + [-ing] → connect

connection → [connect] + [-ion] → connect

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2. Morphological Parsing for Sentiment Analysis

Word 1: unhappy

Rule Applied: Remove prefix **un-**

unhappy

→ un + happy

→ remove un

→ happy

Prefix: un-

Root/Base Form: happy

Type: Derivational

Normalized Output: happy

Word 2: happiness

Rule Applied: Remove suffix **-ness**

happiness
→ happy + ness
→ remove ness
→ happy

Root/Base Form: happy

Suffix: -ness

Type: Derivational

Normalized Output: happy

Word 3: happily

Rule Applied: Remove suffix **-ly**

happily
→ happy + ly
→ remove ly
→ happy

Root/Base Form: happy

Suffix: -ly

Type: Derivational

Normalized Output: happy

Final Morphological Breakdown

unhappy → un + happy → happy
happiness → happy + ness → happy
happily → happy + ly → happy

3. Stemming-Based Preprocessing Module for Search Engine

Word 1: played

Rule Applied: Remove past tense suffix **-ed**

played

→ play + ed

→ remove ed

→ play

Stem: play

Classification: Inflectional

Final Output: play

Word 2: player

Rule Applied: Remove derivational suffix **-er**

player

→ play + er

→ remove er

→ play

Stem: play

Classification: Derivational

Final Output: play

Word 3: playing

Rule Applied: Remove present participle suffix **-ing**

playing

→ play + ing

→ remove ing

→ play

Stem: play

Classification: Inflectional

Final Output: play

Final Stem Extraction

played → play + ed → play

player → play + er → play

playing → play + ing → play

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3. Stemming-Based Preprocessing Module for Search Engine

Word 1: played

Rule Applied: Remove past tense suffix **-ed**

played

→ play + ed

→ remove ed

→ play

Stem: play

Classification: Inflectional

Final Output: play

Word 2: player

Rule Applied: Remove derivational suffix **-er**

player

→ play + er

→ remove er

→ play

Stem: play

Classification: Derivational

Final Output: play

Word 3: playing

Rule Applied: Remove present participle suffix **-ing**

playing

→ play + ing

→ remove ing

→ play

Stem: play

Classification: Inflectional

Final Output: play

Final Stem Extraction

played → play + ed → play

player → play + er → play

playing → play + ing → play

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4. Finite-State Morphological Parsing Module

Word 1: writes

Rule Applied: Add suffix **-s** for third-person singular verb form

write

→ write + s

→ writes

State Transition:

q0(write) → q1(writes)

Root: write

Type: Regular Inflectional

Normalized Output: write

Word 2: writing

Rule Applied: Add suffix **-ing** for present participle form

write

→ write + ing

→ writing

State Transition:

q0(write) → q2(writing)

Root: write

Type: Regular Inflectional

Normalized Output: write

Word 3: written

Rule Applied: Irregular verb transformation

write

→ written

State Transition:

q0(write) → q3(written)

Root: write

Type: Irregular Inflectional

Normalized Output: write

Final Morphological Mapping

writes → write + s → write

writing → write + ing → write

written → irregular form → write

5. Porter Stemmer-Based Preprocessing Module for Document Retrieval

Word 1: relational

Rule Applied: Remove derivational suffix **-al**, then reduce suffix **-ion**

relational

→ relation + al

→ relation

→ relat + ion

→ relat

Intermediate Form: relation

Final Stem: relat

Word 2: relation

Rule Applied: Remove suffix **-ion**

relation

→ relat + ion

→ relat

Intermediate Form: relat

Final Stem: relat

Word 3: relate

Rule Applied: Remove final silent **-e**

relate

→ relat + e

→ relat

Intermediate Form: relat

Final Stem: relat

Final Stemming Output

relational → relation → relat

relation → relat

relate → relat

Rubrics for Calculation

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							

5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____